

Dielectric constant of biological systems

R. Paul

Department of Chemistry, The University of Calgary, Calgary, Alberta, Canada T2N 1N4

J. A. Tuszyński

Department of Physics, Memorial University of Newfoundland, St. John's, Newfoundland, Canada A1B 3X7

R. Chatterjee

Department of Physics, The University of Calgary, Calgary, Alberta, Canada T2N 1N4

(Received 29 November 1983; revised manuscript received 24 February 1984)

In this paper we follow the Fröhlich model of long-range coherence in biological systems and use the field-theoretic Hamiltonian of Wu and Austin. We calculate the complex dielectric constant of the metabolically active dipolar parts in the cell. The actual computation proceeds in two stages. In the first stage the imaginary part of the dielectric constant is obtained by the method of continued fractions. In the second stage the real part of the dielectric constant is calculated by an application of the Kramers-Kronig theorem. The main objective of the work presented in this paper is the real part of the dielectric constant. We also discuss the contributions from water which constitutes the majority of the cell content. Our results confirm the predictions of Fröhlich, and we give analytical expressions which can be utilized in the Fröhlich potential.

I. THE FRÖHLICH MODEL

In a series of papers Fröhlich points out certain physically intriguing features of biological membranes:

- (i) the various components possess a high degree of organization and cooperation;
- (ii) there exists a strong transmembrane electric field;
- (iii) the many types of dipoles present in a membrane are capable of high-frequency vibrations (10^{11} – 10^{12} Hz);
- (iv) in order to support metabolic activity the energy supply must exceed a critical value.

Based on these precepts Fröhlich^{1–7} developed a mathematical model in which a chain of coupled dipole moments interact with their environment consisting of a heat bath and an energy supply. To illustrate the principles involved let us consider just two neighboring dipolar mole-

cules which vibrate initially at frequencies ω_1 and ω_2 , respectively. Then, due to their coupling, modified frequencies of dipole vibrations, ω_+ and ω_- , arise:

$$\omega_{\pm} = \left[\frac{1}{2}(\omega_1^2 + \omega_2^2) \pm \left[\frac{1}{4}(\omega_1^2 - \omega_2^2)^2 + \frac{\beta_0^4}{(\epsilon'_{\pm})^2} \right]^{1/2} \right]^{1/2}, \quad (1)$$

where $\beta_0^2 = \gamma e^2 \sqrt{z_1 z_2} / MR^3$ and γ is a size- and shape-dependent coupling parameter, z_1 and z_2 are the numbers of elementary charges within each of the molecules, R is the equilibrium chain spacing, M is the molecule's mass, and $\epsilon'_{\pm} \equiv \epsilon'(\omega_{\pm})$ is the real part of the complex dielectric constant of the system. The interaction energy

$$\Delta E = \frac{\hbar}{2} [(\omega_+ + \omega_-) - (\omega_1 + \omega_2)] \quad (2)$$

is calculated by expanding the external square root in Eq. (1) as

$$\Delta E \cong \frac{\hbar}{2} \left[2(\omega_1^2 + \omega_2^2)^{1/2} - (\omega_1 + \omega_2) + \frac{\left[(\omega_1^2 - \omega_2^2)^2 + \frac{4\beta_0^4}{(\epsilon'_+)^2} \right]^{1/2} - \left[(\omega_1^2 - \omega_2^2)^2 + \frac{4\beta_0^4}{(\epsilon'_-)^2} \right]^{1/2}}{[8(\omega_1^2 + \omega_2^2)]^{1/2}} \right]. \quad (3)$$

It is only when the two original frequencies are close to resonance, i.e., for $|\omega_1^2 - \omega_2^2| \ll 4\beta_0^2/\epsilon'_{\pm}$, that we can expand the internal square root in a series in which the first term will be proportional to R^{-3} . Otherwise, the suitable expansion contains terms falling off as R^{-6} , R^{-12} , etc. At resonance then ($\omega_1 = \omega_2 = \bar{\omega}$)

$$\Delta E = \frac{A}{R^3} - \frac{B}{R^6}. \quad (4)$$

Here B is the van der Waals constant and A is the long-range constant given by

$$A = \left[\frac{\hbar e^2 z \gamma}{\sqrt{8\bar{\omega} M}} \right] \left[\frac{1}{\epsilon'(\omega_+)} - \frac{1}{\epsilon'(\omega_-)} \right], \quad (5)$$

where $z = z_1 = z_2$. The constant A vanishes when either

- (i) ω_1 and ω_2 differ significantly or

(ii) $\epsilon'(\omega_+) = \epsilon'(\omega_-)$, i.e., the medium is frequency independent in the region considered.

Moreover, A depends on the rate of energy supply S through the value of the resonance frequency $\bar{\omega}$. It is clear that by considering the energy supply time dependent, the harmonic oscillators driven by it will attain the frequency of the driving force. In addition, A is size and shape dependent as a function of γ . If ω_- (ω_+) is excited, then the interaction is a selective attraction (repulsion). All these properties make this mechanism very appealing

$$H = \sum_i \bar{\omega}_i a_i^\dagger a_i + \sum_k \Omega_k b_k^\dagger b_k + \sum_l \theta_l P_l^\dagger P_l + \frac{1}{2} \sum_{i,j,k} (\chi a_i^\dagger a_j b_k^\dagger + \chi^* a_j a_i^\dagger b_k) + \sum_{i,k} (\lambda b_k a_i^\dagger + \lambda^* b_k^\dagger a_i) + \sum_{i,l} (\xi P_l a_i^\dagger + \xi^* P_l^\dagger a_i). \quad (6)$$

Here $(a_i^\dagger, a_i)$, $(b_k^\dagger, b_k)$, and $(P_l^\dagger, P_l)$ are the dipole, heat-bath, and energy-pump creation and annihilation operators, respectively. The first three terms in Eq. (6) describe the three constituent parts of the system in the harmonic approximation and the remaining terms relate to the interactions between the subsystems. The commutation relations for the operators involved in Eq. (6) are those of the Bose-Einstein statistics:

$$[a_i, a_j^\dagger] = [b_i, b_j^\dagger] = [P_i, P_j^\dagger] = \delta_{ij}, \quad (7)$$

$$[a_i, a_j] = [b_i, b_j] = [P_i, P_j] = 0, \quad (8)$$

$$[a_i^\dagger, a_j^\dagger] = [b_i^\dagger, b_j^\dagger] = [P_i^\dagger, P_j^\dagger] = 0. \quad (9)$$

All operators of different types commute. For the present purposes, the Wu-Austin Hamiltonian is approximated by excluding the effects of the pump and linear dipole-heat-bath coupling. These changes, however, can be accommodated by modifying the dipole frequencies from $\bar{\omega}_i$ to ω_i . Following Wu and Austin,¹⁰ the effective Hamiltonian is adopted in the form

$$H = H_0 + H_1, \quad (10)$$

where H_0 is the unperturbed cell Hamiltonian

$$H_0 = \sum_i \omega_i a_i^\dagger a_i \quad (11)$$

and the interaction operator H_1 contains only the dominant, second-order dipole-heat-bath coupling

$$H_1 = \frac{1}{2} \sum_{i,j,k} (\chi a_i^\dagger a_j b_k^\dagger + \chi^* a_j a_i^\dagger b_k). \quad (12)$$

A similar Hamiltonian was used by Mishra *et al.*¹² to calculate the interaction potential between two collections of bosons. The effective Hamiltonian, Eq. (10), will be later used to evaluate the dielectric constant of the biological cell.

III. THE COMPLEX DIELECTRIC CONSTANT

In this section we outline a well-established¹³ approach to the frequency-dependent complex dielectric constant. We have

$$\epsilon(\omega) = \epsilon'(\omega) - i\epsilon''(\omega), \quad (13)$$

from the point of view of growth- and division-control effects in biology.

The only unknown parameter in Eq. (5) has been the dielectric constant $\epsilon'(\bar{\omega})$. This paper is devoted to solving this problem.

II. THE WU-AUSTIN HAMILTONIAN

Wu and Austin⁸⁻¹¹ proposed the following second-quantized Hamiltonian as a model Hamiltonian of the Fröhlich theory:

where $\epsilon'(\omega)$ and $\epsilon''(\omega)$ are the real and imaginary parts, respectively. The Kramers-Kronig¹⁴ theorem relates the two parts through

$$\epsilon'(\omega) = 1 + \frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{\epsilon''(\nu)}{\nu - \omega} d\nu \quad (14)$$

and

$$\epsilon''(\omega) = -\frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{\epsilon'(\nu)}{\nu - \omega} d\nu, \quad (15)$$

where $\epsilon_\infty = \epsilon(\infty)$ and P denotes taking the principal part of the integral. Moreover, the imaginary part of $\epsilon(\omega)$ is proportional to the band-shape function $I(\omega)$ which is an observable quantity

$$\epsilon''(\omega) = \frac{4\pi^2(1 - e^{-\beta\hbar\omega})}{V\hbar} I(\omega), \quad (16)$$

where V is the volume, $\hbar$ is Planck's constant, and $\beta = (kT)^{-1}$ with k the Boltzmann constant and T the absolute temperature. The band-shape function, in turn, is derived from the dipole-dipole autocorrelation function by the Fourier transformation

$$I(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-i\omega t} \langle \hat{\mathbf{M}} \cdot \hat{\mathbf{M}}(t) \rangle dt. \quad (17)$$

Here, $\hat{\mathbf{M}}$ is the total-dipole-moment operator of the sample:

$$\hat{\mathbf{M}} = \sum_{i=1}^N \hat{\boldsymbol{\mu}}_i \quad (18)$$

and t is time. The angular brackets $\langle \rangle$ imply a thermal average of the operators within, i.e.,

$$\langle \hat{\mathbf{M}} \cdot \hat{\mathbf{M}}(t) \rangle \equiv \text{Tr} \left[\left[\frac{e^{-\beta H}}{Z} \right] \hat{\mathbf{M}} \cdot \hat{\mathbf{M}}(t) \right], \quad (19)$$

where H is the Hamiltonian of the system and Z is the canonical partition function. The time development of the operator $\hat{\mathbf{M}}(t)$ is given by

$$\hat{\mathbf{M}}(t) = e^{i\hat{L}t} \hat{\mathbf{M}}, \quad (20)$$

where $\hat{L}$ is the Liouville operator defined by

$$\hat{L}\hat{M} \equiv \frac{1}{\hbar} [H, \hat{M}] . \quad (21)$$

Furthermore, the average $\langle \hat{M} \cdot \hat{M}(t) \rangle$ is related to the causal Green's function $\bar{G}(t)$ since

$$\bar{G}(t) \equiv -i\Theta(t) \langle \hat{M}(t) \cdot \hat{M} \rangle - i\Theta(-t) \langle \hat{M} \cdot \hat{M}(t) \rangle , \quad (22)$$

where $\Theta(t)$ is the Heaviside step function. Denoting the Fourier transform of $\bar{G}(t)$ by $\bar{G}(\omega)$ we arrive at

$$I(\omega) = \frac{-1}{1 - e^{-\beta\hbar\omega}} \text{Im} \bar{G}(\omega) . \quad (23)$$

Defining the binary product of two arbitrary operators $\hat{B}$ and $\hat{C}$ as

$$(\hat{B}^\dagger | \hat{C}) \equiv \text{Tr} \left[\left[\frac{e^{-\beta H}}{Z} \right] [\hat{B}, \hat{C}] \right] \quad (24)$$

and employing the complex frequency variable $z = i\omega \pm \eta$, $\bar{G}(z)$ acquires a very compact form:

$$\frac{2\pi}{i} \bar{G}(z) = \left[\hat{M}^\dagger \left| \frac{1}{z - i\hat{L}} \right| \hat{M} \right] . \quad (25)$$

The operator $\hat{M}$ can be written in a second-quantized form as

$$\hat{M} = \sum_{j,l} \langle j | \hat{\mu} | l \rangle a_j^\dagger a_l , \quad (26)$$

where the creation and annihilation operators $a_j^\dagger$ and a_l span the complete and orthonormal space of many-body states $|n_1, n_2, \dots\rangle$:

$$a_i^\dagger |n_1, n_2, \dots, n_i, \dots\rangle = \sqrt{n_i + 1} |n_1, n_2, \dots, n_i + 1, \dots\rangle , \quad (27)$$

$$a_i |n_1, n_2, \dots, n_i, \dots\rangle = \sqrt{n_i} |n_1, n_2, \dots, n_i - 1, \dots\rangle , \quad (28)$$

where n_i is the occupation number in the i th state ϕ_i . Substituting Eq. (26) into Eq. (25) yields

$$\bar{G}(z) = \sum_{j,l,m,n} \langle j | \hat{\mu} | l \rangle \langle m | \hat{\mu} | n \rangle G_{jlmn}(z) , \quad (29)$$

where $G_{jlmn}(z)$ is a Green's function in terms of the field operators alone,

$$\frac{2\pi}{i} G_{jlmn}(z) \equiv \left[F_{jl} \left| \frac{1}{z - i\hat{L}} \right| F_{mn} \right] , \quad (30)$$

where $F_{ij} \equiv a_i^\dagger a_j$. Equation (30) can be equivalently written in a matrix notation as

$$\frac{2\pi}{i} \underline{G}(z) = \left[\underline{F} \left| \frac{1}{z - i\hat{L}} \right| \underline{F} \right] . \quad (31)$$

By introducing a projection operator P ,

$$P \equiv |\underline{F}\rangle \underline{\alpha} \langle \underline{F}| , \quad (32)$$

where $\underline{\alpha} = (\underline{F} | \underline{F})^{-1}$ and P 's complement $Q \equiv 1 - P$ and

following the prescription due to Mori,¹⁵ the Green's function, Eq. (31), can be converted into a continued fraction expression

$$\begin{aligned} \frac{2\pi}{i} \underline{G}(z) &= \frac{(\underline{F} | \underline{F})}{z \underline{1} - (\underline{F} | i\hat{L} | \underline{F}) \cdot \underline{\alpha} + \frac{(\underline{G} | \underline{G}) \cdot \underline{\alpha}}{z \underline{1} - (\underline{G} | i\hat{L}_1 | \underline{G}) \cdot \underline{\beta} + \dots}} , \end{aligned} \quad (33)$$

where the following symbols have been used:

$$| \underline{G} \rangle \equiv Q i\hat{L} | \underline{F} \rangle , \quad (34)$$

$$\hat{L}_1 \equiv Q \hat{L} Q , \quad (35)$$

$$\underline{\beta} \equiv (\underline{G} | \underline{G})^{-1} , \quad (36)$$

and $\underline{1}$ is an identity matrix.

Equation (33) is the central equation of this paper. Its Fourier transform will be used to calculate the band-shape function $I(\omega)$ via Eq. (23). This, in turn, will enable us to evaluate both imaginary and real parts of the dielectric constant, using Eqs. (16) and (14), respectively. However, the continued fraction expression, Eq. (33), is just another way of formally representing the Green's function $\underline{G}(z)$. It is an exact result which, unless further simplifications are given, must not be viewed as a solution to the problem. In Sec. IV we employ an effective Hamiltonian, Eq. (10), and carry out approximate calculations of the continued fraction expression, Eq. (33).

IV. APPROXIMATE CALCULATIONS OF THE GREEN'S FUNCTION

In order to calculate the Green's function of the Hamiltonian, Eq. (10), we rewrite Eq. (33) as

$$\frac{2\pi}{i} \underline{G}(z) = [z \underline{1} - (\underline{F} | i\hat{L} | \underline{F}) \cdot \underline{\alpha} - \underline{\sigma}(z)]^{-1} \cdot (\underline{F} | \underline{F}) , \quad (37)$$

where

$$\underline{\sigma}(z) \equiv - \left[\underline{G} \left| \frac{1}{z - i\hat{L}_1} \right| \underline{G} \right] \cdot \underline{\alpha} \quad (38)$$

and $\underline{\sigma}(z)$ must be approximated. The details of our calculations involved in Eq. (37) are presented in Appendix A and here we just summarize the results of subsequent stages of these calculations.

(i) Matrix $(\underline{F} | \underline{F})$ is given by

$$(F_{jl} | F_{mn}) = (n_l - n_m) \delta_{j,m} \delta_{l,n} , \quad (39)$$

where n_i denotes the number of dipolar phonons in the state i

$$n_i = [\exp(\omega_i/kT) - 1]^{-1} \quad (40)$$

according to the Bose-Einstein statistics.

(ii) Matrix $\underline{\alpha}$ is the inverse of $(\underline{F} | \underline{F})$. Since the latter matrix is diagonal, so is $\underline{\alpha}$:

$$\alpha_{jlmn} = (n_l - n_m)^{-1} \delta_{j,m} \delta_{l,n} . \quad (41)$$

(iii) Matrix $(\underline{F} | i\hat{L} | \underline{F})$ contains the following elements:

$$(F_{jl} | i\hat{L} | F_{mn}) = \frac{i}{\hbar} (\omega_m - \omega_n) (n_l - n_m) \delta_{j,m} \delta_{l,n} . \quad (42)$$

(iv) Matrix $(\underline{G} | (z - i\hat{L}_1)^{-1} | \underline{G})$ equals $(\underline{F} | i\hat{L}Q \times (z - i\hat{L}_1)^{-1} Q i\hat{L} | \underline{F})$. However, its elements cannot be obtained exactly. The approximate calculations carried out in Appendix A yield

$$\begin{aligned} \left[F_{jl} \left| i\hat{L}Q \frac{1}{z - i\hat{L}_1} Q i\hat{L} \right| F_{mn} \right] = \frac{|\chi|^2}{\hbar} & \left[\sum_{p,r} \frac{\rho_r(n_l - n_p) - \bar{n}_l n_p}{z + i\hbar^{-1}\omega_{p,l} - i\hbar^{-1}\Omega_r} \delta_{l,n} - \sum_r \frac{\rho_r(n_l - n_m) - \bar{n}_l n_m}{z + i\hbar^{-1}\omega_{m,l} - i\hbar^{-1}\Omega_r} \right. \\ & - \sum_r \frac{\rho_r(n_n - n_j) - \bar{n}_n n_j}{z + i\hbar^{-1}\omega_{j,n} - i\hbar^{-1}\Omega_r} + \sum_{p,r} \frac{\rho_r(n_p - n_m) - \bar{n}_p n_m}{z + i\hbar^{-1}\omega_{m,p} - i\hbar^{-1}\Omega_r} \delta_{j,m} \\ & + \sum_{p,r} \frac{\bar{\rho}_r(n_l - n_p) - \bar{n}_l n_p}{z + i\hbar^{-1}\omega_{p,l} + i\hbar^{-1}\Omega_r} \delta_{l,n} - \sum_r \frac{\bar{\rho}_r(n_l - n_m) + \bar{n}_l n_m}{z + i\hbar^{-1}\omega_{m,l} + i\hbar^{-1}\Omega_r} \\ & \left. - \sum_r \frac{\bar{\rho}_r(n_n - n_j) + \bar{n}_n n_j}{z + i\hbar^{-1}\omega_{j,n} + i\hbar^{-1}\Omega_r} + \sum_{p,r} \frac{\rho_r(n_p - n_m) + \bar{n}_p n_m}{z + i\hbar^{-1}\omega_{m,p} + i\hbar^{-1}\Omega_r} \delta_{j,m} \right] . \quad (43) \end{aligned}$$

The following symbols have been used:

$$\omega_{i,j} \equiv \omega_i - \omega_j , \quad (44)$$

$$\bar{n}_i \equiv 1 + n_i , \quad (45)$$

$$\rho_i \equiv [\exp(\Omega_i/kT) - 1]^{-1} , \quad (46)$$

$$\bar{\rho}_i \equiv 1 + \rho_i . \quad (47)$$

At this point we have calculated all the matrices that are required in Eq. (37), but we still must carry out a matrix inversion. This inversion is preceded by deriving a Dyson equation which is then solved by a variational technique. The details of this computation are given in Appendix B. Based on this, the approximate Green's function is

$$\begin{aligned} \frac{2\pi}{i} G_{jlmn}(z) = & \left[z - i\hbar^{-1}\omega_{j,l} - \frac{|\chi|^2}{\hbar^2(n_m - n_n)} \left[\sum_{p,r} \frac{\rho_r(n_l - n_p) - \bar{n}_l n_p}{z + i\hbar^{-1}\omega_{p,l} - i\hbar^{-1}\Omega_r} - \sum_p \frac{\bar{\rho}_p(n_l - n_m) - \bar{n}_l n_m}{z + i\hbar^{-1}\omega_{m,l} - i\hbar^{-1}\Omega_p} \right. \right. \\ & - \sum_p \frac{\bar{\rho}_p(n_n - n_j) - \bar{n}_n n_j}{z + i\hbar^{-1}\omega_{j,n} - i\hbar^{-1}\Omega_p} + \sum_{p,r} \frac{\rho_r(n_p - n_m) - \bar{n}_p n_m}{z + i\hbar^{-1}\omega_{m,p} - i\hbar^{-1}\Omega_r} \\ & + \sum_{p,r} \frac{\rho_r(n_l - n_p) - \bar{n}_l n_p}{z + i\hbar^{-1}\omega_{p,l} + i\hbar^{-1}\Omega_r} - \sum_p \frac{\bar{\rho}_p(n_l - n_m) + \bar{n}_l n_m}{z + i\hbar^{-1}\omega_{m,l} + i\hbar^{-1}\Omega_p} \\ & - \sum_p \frac{\bar{\rho}_p(n_n - n_j) + \bar{n}_n n_j}{z + i\hbar^{-1}\omega_{j,n} + i\hbar^{-1}\Omega_p} \\ & \left. \left. + \sum_{p,r} \frac{\rho_r(n_p - n_m) + \bar{n}_p n_m}{z + i\hbar^{-1}\omega_{m,p} + i\hbar^{-1}\Omega_r} \right] \right]^{-1} (n_n - n_m) \delta_{j,m} \delta_{l,n} . \quad (48) \end{aligned}$$

This complicated equation can be written in a compact form:

$$\frac{2\pi}{i} G_{jlmn}(z) \equiv \left[z - i\hbar^{-1}\omega_{j,l} - \frac{|\chi|^2}{\hbar^2(n_m - n_n)} \sum_{Q=1}^8 \sigma_{jlmn}^{(Q)}(z) \right]^{-1} (n_l - n_j) \delta_{j,m} \delta_{l,n} . \quad (49)$$

The precise definition of the eight quantities $\sigma_{jlmn}^{(Q)}(z)$ can be obtained by comparing the right-hand sides of Eqs. (48) and (49).

The Green's function given by Eq. (49) depends upon the complex frequency variable z . The corresponding function dependent upon the real frequency ω is obtained by the method outlined in Appendix C. The final form of the required Green's function is

$$G_{jlmn}(\omega) = \frac{1}{2\pi} \frac{(n_l - n_j) \delta_{j,m} \delta_{l,n}}{\omega - \hbar^{-1}\omega_{j,l} \pm [i|\chi|^2/\pi\hbar^2(n_l - n_j)] \Gamma_{jl}(\omega)} . \quad (50)$$

The function $\Gamma_{jl}(\omega)$ is the frequency-dependent linewidth function. Note that the line shift function $S(\omega)$ has been

neglected since the resonance frequencies are usually determined from experiment anyhow. The form of the linewidth function will be discussed in Sec. V.

V. CALCULATIONS OF THE COMPLEX DIELECTRIC CONSTANT

Having found the explicit form of the frequency-dependent correlation function, Eq. (50), we use Eqs. (23) and (16) to obtain the imaginary (absorptive) part of $\epsilon(\omega)$ and then, applying the Kramers-Kronig formula, Eq. (14), the real (nonabsorptive) part of $\epsilon(\omega)$ will be calculated. We start with the imaginary part of $G_{jlmn}(\omega)$,

$$\text{Im } G_{jlmn}(\omega) = \frac{1}{2\pi} \frac{(|\chi|^2/\pi\hbar^2)\Gamma_{jl}(\omega)\delta_{j,m}\delta_{l,n}}{(\omega - \hbar^{-1}\omega_{j,l})^2 + [|\chi|^4/\pi^2\hbar^4(n_l - n_j)^2]\Gamma_{jl}^2(\omega)} . \quad (51)$$

Assuming the dipole moment matrix elements in the phonon states in Eq. (26), $\langle i | \hat{\mu} | j \rangle$, to be real quantities, we obtain

$$\begin{aligned} \text{Im } \bar{G}(\omega) &= \sum_{j,l,m,n} \langle j | \hat{\mu} | l \rangle \langle m | \hat{\mu} | n \rangle \text{Im } G_{jlmn}(\omega) \\ &= \sum_{j,l} |\langle j | \hat{\mu} | l \rangle|^2 \frac{(|\chi|^2/\pi\hbar^2)\Gamma_{jl}(\omega)}{(\omega - \hbar^{-1}\omega_{j,l})^2 + [|\chi|^4/\pi^2\hbar^4(n_l - n_j)^2]\Gamma_{jl}^2(\omega)} . \end{aligned} \quad (52)$$

This, together with Eqs. (23) and (16), is used to evaluate the imaginary part of the dielectric constant:

$$\epsilon''(\omega) = \frac{4\pi^2(1 - e^{-\beta\hbar\omega})}{V\hbar(1 + e^{-\beta\hbar\omega})} \sum_{j,l} |\langle j | \hat{\mu} | l \rangle|^2 \frac{(|\chi|^2/\pi\hbar^2)\Gamma_{jl}(\omega)}{(\omega - \hbar^{-1}\omega_{j,l})^2 + [|\chi|^4/\pi^2\hbar^4(n_l - n_j)^2]\Gamma_{jl}^2(\omega)} . \quad (53)$$

If the imaginary part of the dielectric constant is to have any practical value, an explicit form for the quantity $\Gamma_{jl}(\omega)$ must be obtained. The mathematical details have been relegated to Appendix D and we outline only the most important points here.

(i) If, in Eq. (48), the coupling constant χ between the dipolar and the bath phonons is allowed to vanish, a very simple formula is obtained:

$$\frac{2\pi}{i} G_{jlmn}(z) \cong \frac{(n_n - n_m)\delta_{j,m}\delta_{l,n}}{z - i\hbar^{-1}\omega_{j,l}} . \quad (54)$$

Now, applying the principles discussed in Appendix C, we find

$$\text{Im } G_{jlmn}(\omega) = \pi(n_n - n_m)\hbar\delta_{j,m}\delta_{l,n}\delta(\hbar\omega - (\omega_j - \omega_l)) . \quad (55)$$

Thus the absorptive part of the dielectric constant calculated from this equation displays a sharp (narrow) line corresponding to a dipolar frequency transition given by $\omega_l \rightarrow \omega_j$, where $\omega_j > \omega_l$. An absorption line will be observed at an experimental frequency $\omega = \hbar^{-1}(\omega_j - \omega_l)$.

(ii) As the effects of the bath phonons are introduced ($\chi \neq 0$), the situation described in (i) is altered as both shifting and broadening of the absorption line is observed. Focusing on the transitions between the modes j and l , two types of terms in Eq. (48) can be distinguished.

(a) The bath directly influences the modes j and l :

$$\text{Bath} \rightarrow \begin{cases} \text{mode } j \\ \text{mode } l \end{cases} .$$

This coupling scheme corresponds to those terms in Eq. (48) that contain a single summation over p .

(b) The bath influences a dipolar mode other than the modes j and l and the result of this is fed into the modes j and l :

$$\text{Bath} \rightarrow \text{intermediate dipolar mode} \rightarrow \begin{cases} \text{mode } j \\ \text{mode } l \end{cases} .$$

This coupling scheme corresponds to all those terms in Eq. (48) that contain a double summation over p and r .

(iii) The summations involved in terms of category (a) can be easily carried out but some difficulty is associated with the terms of category (b). To simplify, it is assumed that there exists only one dipolar mode k other than modes j and l participating in processes (b) and that the bath phonons are acoustic phonons with sound velocity v . The function $\Gamma_{jl}(\omega)$ is now evaluated (see Appendix D for details) as

$$\begin{aligned} \Gamma_{jl}(\omega) &= \frac{V\hbar}{2\pi^2v^3} \{ w_1^2(n_l - n_k)[\rho(w_1) + \rho(-w_1)] \\ &\quad + w_2^2(n_k - n_j)[\rho(w_2) + \rho(-w_2)] \\ &\quad - 2\bar{w}_1^2(n_l - n_j)[\bar{\rho}(\bar{w}_1) + \bar{\rho}(-\bar{w}_1)] \} , \end{aligned} \quad (56)$$

where

$$w_1 \equiv \hbar\omega + \omega_{k,l} , \quad (57)$$

$$w_2 \equiv \hbar\omega + \omega_{j,k} , \quad (58)$$

$$\bar{w}_1 \equiv \hbar\omega + \omega_{j,l} . \quad (59)$$

We now calculate the real part of the dielectric constant using the Kramers-Kronig formula, Eq. (14), and the expression for $\epsilon''(\omega)$, Eq. (53). In order to simplify the result we make the following two approximations:

- (i) the resonance approximation, $\omega \cong \hbar^{-1}\omega_{j,l}$, and
- (ii) the high-temperature limit, $kT \gg \hbar\omega$.

The necessary integration is carried out with the aid of a contour in the complex plane (see Appendix E). The final expression for the real part of the dielectric constant is obtained as

$$\epsilon'(\omega) = 1 + \frac{2\pi}{VkT} \sum_{j,l} |\langle j | \hat{\mu} | l \rangle|^2 (n_l - n_j) \frac{\left[\hbar^{-2} \omega_{j,l}^2 + \left(\frac{|\chi|^4}{\pi^2 \hbar^4 (n_l - n_j)^2} \right) \Gamma_{jl}^2(\hbar^{-1} \omega_{j,l}) - \hbar^{-1} \omega_{j,l} \omega \right]}{\left[(\omega - \hbar^{-1} \omega_{j,l})^2 + \frac{|\chi|^4}{\pi^2 \hbar^4 (n_l - n_j)^2} \Gamma_{jl}^2(\hbar^{-1} \omega_{j,l}) \right]}. \quad (60)$$

VI. DISCUSSION

In our Hamiltonian the coupling constant χ is an unknown parameter. The equations that have been derived for ϵ' and ϵ'' , which are measurable quantities, should provide a means for estimating this parameter. Since both Eqs. (53) and (60) are fairly complicated, we introduce two additional assumptions:

- (i) the resonance condition, $\omega = \omega_0 \cong \hbar^{-1} \omega_{j,l}$, and
- (ii) the high-temperature limit, $\exp(-\beta \hbar \omega) \cong 1 - \beta \hbar \omega$.

As a result, an approximate expression for $\epsilon''(\omega)$ is obtained:

$$\epsilon''(\omega_0) \cong \frac{2\pi^3 \hbar^2 \omega_0}{VkT |\chi|^2} \sum_{j,l} \frac{|\langle j | \hat{\mu} | l \rangle|^2}{\Gamma_{jl}(\omega_0)} (n_l - n_j)^2. \quad (61)$$

Furthermore, we approximate the remaining summation by

$$\sum_{j,l} \frac{|\langle j | \hat{\mu} | l \rangle|^2}{\Gamma_{jl}(\omega_0)} (n_l - n_j)^2 \cong \frac{N |\hat{\mu}|^2}{\Gamma(\omega_0)}, \quad (62)$$

where N is the total number of phonons and $|\hat{\mu}|$ is the mean dipole moment of the sample. Hence, the final form of $\epsilon''(\omega)$ is

$$\epsilon''(\omega_0) \cong \frac{2\pi^3 \hbar^2 \rho |\hat{\mu}|^2 \omega_0}{|\chi|^2 kT \Gamma(\omega_0)}, \quad (63)$$

where ρ is the density of the material.

Similarly, the real part of the dielectric constant, $\epsilon'(\omega)$ of Eq. (60), can be evaluated close to the resonance, i.e., for $\omega = \omega_0 \pm \delta\omega$, as

$$\epsilon'(\omega_0 \pm \delta\omega) = 1 + \frac{2\pi}{kT} \rho |\hat{\mu}|^2 \frac{1 \mp \omega_0 \Delta \delta\omega}{1 - \Delta(\delta\omega)^2}, \quad (64)$$

where $\Delta \cong \pi^2 \hbar^4 N / |\chi|^4 \Gamma^2(\omega_0)$. Therefore, $\epsilon'(\omega)$ is of an asymmetric shape whose value at ω_0 is equal to the static value

$$\epsilon'(\omega_0) = 1 + \frac{2\pi}{kT} \rho |\hat{\mu}|^2 = \epsilon'(0) \quad (65)$$

and $\epsilon'(\pm\infty) = 1$. Equation (65) resembles the Debye equation in the case of ϵ resulting from the existence of permanent dipole moments. Within this approximation the real part, $\epsilon'(\omega_0)$, is strongly frequency dependent, confirming the assumption of Fröhlich. We can therefore conclude that the point dipoles do contribute to the long-range attraction predicted by Fröhlich. The imaginary part, $\epsilon''(\omega_0)$, is proportional to ω_0 and inversely proportional to the coupling constant $|\chi|^2$. The real part becomes frequency independent as $|\chi| \rightarrow 0$. Both ϵ' and ϵ'' decrease with the absolute temperature and increase with

the density of the substance.

Our calculations of the complex dielectric constant of a biological cell have been based on a number of assumptions which have to be scrutinized.

(i) The omission of energy-pump and linear cell-heat-bath coupling can be justified as causing a constant shift of eigenfrequencies $\omega_i \rightarrow \tilde{\omega}_i$ which are not known beforehand.

(ii) Similar argument applies to neglecting the line-shift function $S(\omega)$.

(iii) The continued fraction expression, Eq. (48), is approximate. However, it is still complicated enough for our purposes.

(iv) In dealing with the heat-bath modes we have adopted continuum approximation and Debye model, both realistic in view of the large size of the heat bath.

(v) The high-temperature limit seems appropriate as $T \cong 300$ K is a typical temperature necessary for life processes.

(vi) Finally, taking only one excited mode k into consideration in Eq. (56) is yet to be verified but it seems compatible with the tenets of the Fröhlich model.

It should be realized that the derived complex dielectric constant, Eqs. (53) and (60), covers the giant dipoles present in the cell membrane. It does not include the bulk of a cell which is free and bound water. This fact must be accounted for when attempting to correlate our theory with experiment. Unfortunately, no satisfactory theory on the role of water content in the dielectric properties of organic materials exists. It is well known¹⁶ that water affects the value of ϵ'' more profoundly than ϵ' . For pure water ϵ'' reaches 30 while dry organic material has ϵ'' in the range 10^{-3} – 10^{-1} (microwave frequencies considered). Since moist materials achieve ϵ'' in the order of 1, both water and organic molecules seem to be involved strongly in dielectric processes.

Based on the excellent treatise on aqueous dielectrics by Hasted,¹⁶ we write the expression for the dielectric constant of an entire tissue approximately as

$$\epsilon = v_m \epsilon_m + v_b \epsilon_b + (1 - v_m - v_b) \epsilon_w, \quad (66)$$

where v_m and v_b are the volume fractions of the membrane dipolar complexes and bound water, respectively, and ϵ_m, ϵ_b are the corresponding dielectric constants. The values of v_m and v_b vary from substance to substance, but for muscle tissue,¹⁶ for example, $v_m = 25\%$. The dielectric constant of free water ϵ_w can be assumed to obey the Debye-Drude equations

$$\epsilon'(\omega) = \epsilon_\infty + \frac{\epsilon_s - \epsilon_\infty}{1 + \omega^2 \tau^2}, \quad (67)$$

$$\epsilon''(\omega) = \frac{(\epsilon_s - \epsilon_\infty) \omega \tau}{1 + \omega^2 \tau^2}, \quad (68)$$

where $\epsilon_s = \epsilon(0)$, $\epsilon_\infty = \epsilon(\infty)$, and τ is the single relaxation time. At 20°C, the most appropriate temperature for considering living organisms, the free-water parameters, are¹⁶

$$\epsilon_s \cong 78.5, \quad \epsilon_\infty \cong 4.5, \quad \tau = 0.9 \times 10^{-11} \text{ s}.$$

The dipole moment of H₂O molecules is $|\vec{\mu}| = 1.84 \text{ D}$ whereas for various protein molecules it ranges between 150 and 400 D depending on the type of the protein and the pH of the electrolyte.

The dielectric properties of bound water are not known in detail but are presumed to be between those of ice and liquid water. Its volume fraction v_b is much smaller than that of free water and it relaxes at much lower frequencies with almost no contribution to the dielectric loss at microwave frequencies.

Finally, the dielectric constant of the membrane dipolar complexes ϵ_m is given in Eqs. (53) and (60) or their estimates, Eqs. (63) and (65). We believe that it should be possible to experimentally detect these unusual dielectric properties of biological substances. We expect that in the 10^{10} – 10^{12} Hz region there should be resonance peaks clearly distinguishable from the water background. After all, the membrane content in the tissue is by no means negligible. At this stage of development we are not prepared to offer any numerical values, which is simply due to the scarcity of relevant experimental data. Nevertheless we recognize the importance of this question and will deal with it in our future work.

APPENDIX A

(i) In order to calculate the matrix elements of $(\hat{F} | \hat{F})$ we evaluate the commutator

$$\begin{aligned} i\hat{L}_1 = i\hat{L} - \sum_{s,t} i\hat{L} | a_s^\dagger a_t \rangle \frac{1}{n_t - n_s} (a_s^\dagger a_t | - \sum_{q,r} | a_q^\dagger a_r \rangle \frac{1}{(n_r - n_q)} (a_q^\dagger a_r | i\hat{L} \\ + \sum_{q,r,s,t} | a_q^\dagger a_r \rangle \frac{1}{(n_r - n_q)} (a_q^\dagger a_r | i\hat{L} | a_s^\dagger a_t \rangle \frac{1}{(n_t - n_s)} (a_s^\dagger a_t | . \end{aligned} \quad (\text{A7})$$

(d) Making use of Eq. (A6) we obtain

$$\left(G_{jl} \left| \frac{1}{z - i\hat{L}_1} \right| G_{mn} \right) = \left[-\frac{i}{\hbar} \right] \left[\frac{i}{\hbar} \right] \sum_{p,q,r,s} \left[(\chi^* b_s + \chi b_s^\dagger)(F_{pl} - F_{jr}) \left| \frac{1}{z - i\hat{L}_1} \right| (\chi^* b_q + \chi b_q^\dagger)(F_{pn} - F_{mq}) \right] \quad (\text{A8})$$

and upon expanding the right-hand side, eight terms arise, each of which is of the form

$$\left(b_l F_{jk} \left| \frac{1}{z - i\hat{L}_1} \right| b_l F_{mn} \right) \equiv G_{ijklmn}^{(1)}(z), \quad (\text{A9})$$

$$\left(b_l^\dagger F_{jk} \left| \frac{1}{z - i\hat{L}_1} \right| b_l^\dagger F_{mn} \right) \equiv G_{ijklmn}^{(2)}(z). \quad (\text{A10})$$

These new Green's functions involve the new variables bF and $b^\dagger F$ which implies that as we proceed further down the continued fraction, the bath variables begin to make their appearance. It is convenient to introduce the

$$[F_{ik}, F_{jl}] = \delta_{kj} a_i^\dagger a_l - \delta_{il} a_j^\dagger a_k \quad (\text{A1})$$

and use Eq. (24) to find

$$(F_{jl} | F_{mn}) = (n_l - n_m) \delta_{j,m} \delta_{l,n}, \quad (\text{A2})$$

where we have approximated H by H_0 .

(ii) The calculations of $(\hat{F} | i\hat{L} | \hat{F})$ require applying the Liouville operator $\hat{L}$ of Eq. (21) to the element F_{mn} :

$$\begin{aligned} i\hat{L} F_{mn} = \frac{i}{\hbar} \left[(\omega_m - \omega_n) a_m^\dagger a_n + \sum_{p,r} \chi^* (a_p^\dagger a_n - a_n^\dagger a_p) b_r \right. \\ \left. + \sum_{p,r} \chi (a_p^\dagger a_n - a_n^\dagger a_p) b_r^\dagger \right]. \end{aligned} \quad (\text{A3})$$

Taking the inner product of the "ket" produced from this result with the "bra" $(F_{jl} |$ yields

$$(F_{jl} | i\hat{L} | F_{mn}) = \frac{i}{\hbar} (\omega_m - \omega)(n_l - n_m) \delta_{j,m} \delta_{l,n}. \quad (\text{A4})$$

(iii) The matrix $(\underline{G} | (z - i\hat{L}_1)^{-1} | \underline{G})$ is calculated in several steps.

(a) Using $Q = 1 - P$ and Eq. (32) for P , we write Q explicitly as

$$Q = 1 - \sum_{q,r} | a_q^\dagger a_r \rangle \frac{1}{(n_r - n_q)} (a_q^\dagger a_r | . \quad (\text{A5})$$

(b) The definition of $|\underline{G}\rangle$, Eq. (34), applied in conjunction with Eq. (A5) gives

$$| G_{mn} \rangle = \frac{i}{\hbar} \sum_{p,r} | (\chi^* b_r + \chi b_r^\dagger)(F_{pn} - F_{mp}) \rangle. \quad (\text{A6})$$

(c) Similarly, utilizing Eq. (A5) in the definition of $\hat{L}_1$, Eq. (35) yields

vectors $\underline{A}$ and $\underline{B}$ of which bF and $b^\dagger F$ are the components. Thus we have $b_l F_{jk} \in \underline{A}$ and $b_l^\dagger F_{jk} \in \underline{B}$. Hence the right-hand sides of Eqs. (A9) and (A10) are the elements of the following matrices: $G_{ijklmn}^{(1)}(z) \in \underline{G}^{(1)}(z)$, $G_{ijklmn}^{(2)}(z) \in \underline{G}^{(2)}(z)$. Formally, these new Green's functions resemble Eq. (37) since

$$\underline{G}^{(1)}(z) = [z \underline{1} - (\underline{A} | i\hat{L}_1 | \underline{A}) - \underline{\Sigma}^{(1)}(z)]^{-1} \cdot (\underline{A} | \underline{A}) \quad (\text{A11})$$

and

$$\underline{G}^{(2)}(z) = [z \underline{1} - (\underline{B} | i\hat{L}_1 | \underline{B}) - \underline{\Sigma}^{(2)}(z)]^{-1} \cdot (\underline{B} | \underline{B}). \quad (\text{A12})$$

Here, the two functions $\underline{\Sigma}^{(1)}(z)$ and $\underline{\Sigma}^{(2)}(z)$ represent the third stage in the continued fraction development and are set equal to zero as an approximation:

$$\underline{\Sigma}^{(1)}(z) = \underline{\Sigma}^{(2)}(z) = 0. \quad (\text{A13})$$

With this approximation Eqs. (A11) and (A12) may be written as

$$\underline{G}^{(1)}(z) \simeq (\underline{A} | \underline{A}) \cdot [\underline{\Delta}^{(1)}(z)]^{-1} \cdot (\underline{A} | \underline{A}), \quad (\text{A14})$$

$$\underline{G}^{(2)}(z) \simeq (\underline{B} | \underline{B}) \cdot [\underline{\Delta}^{(2)}(z)]^{-1} \cdot (\underline{B} | \underline{B}), \quad (\text{A15})$$

where

$$\underline{\Delta}^{(1)}(z) \equiv [z(\underline{A} | \underline{A}) - (\underline{A} | i\hat{L}_1 | \underline{A})], \quad (\text{A16})$$

$$\underline{\Delta}^{(2)}(z) \equiv [z(\underline{B} | \underline{B}) - (\underline{B} | i\hat{L}_1 | \underline{B})]. \quad (\text{A17})$$

The final forms of the matrix elements are written as follows:

$$G_{ijklmn}^{(1)}(z) = \frac{\rho_i(n_k - n_m) - \bar{n}_k n_m}{z + i\hbar^{-1}\omega_{m,n} - i\hbar^{-1}\Omega_1} \delta_{il} \delta_{jm} \delta_{kn}, \quad (\text{A18})$$

$$G_{ijklmn}^{(2)}(z) = \frac{\bar{\rho}_i(n_k - n_m) + \bar{n}_k n_m}{z + i\hbar^{-1}\omega_{m,n} + i\hbar^{-1}\Omega_1} \delta_{il} \delta_{jm} \delta_{kn}. \quad (\text{A19})$$

The various quantities appearing in these equations are defined in Sec. IV [see Eqs. (40) and (44)–(47)].

APPENDIX B

In this appendix we calculate the inverse required in Eq. (37) using a variational technique.¹⁵ For convenience we extract from Eq. (37) the part that has to be inverted and represent it by the matrix $\underline{g}(z)$,

$$\underline{g}(z) \equiv [z\underline{1} - (F | i\hat{L} | F) \cdot \underline{\alpha} - \underline{\sigma}(z)]^{-1}. \quad (\text{B1})$$

We first convert Eq. (B1) into a Dyson equation defining a zero-order diagonal Green's function:

$$\underline{g}_0^{-1}(z) = z\underline{1} - (F | i\hat{L} | F) \cdot \underline{\alpha}. \quad (\text{B2})$$

Therefore

$$\underline{g}^{-1}(z) = \underline{g}_0^{-1}(z) - \underline{\sigma}(z). \quad (\text{B3})$$

Multiplying both sides of this equation by $\underline{g}(z)$ and then from the left by $\underline{g}_0(z)$ and rearranging we obtain the desired Dyson equation

$$\underline{g}(z) = \underline{g}_0(z) + \underline{g}_0(z) \cdot \underline{\sigma}(z) \cdot \underline{g}(z). \quad (\text{B4})$$

Since $\underline{\sigma}$ is large we use a variational approach to solve the Dyson equation instead of the iterative method which employs

$$\begin{aligned} \underline{g}(z) &= \underline{g}_0(z) + \underline{g}_0(z) \cdot \underline{\sigma}(z) \cdot \underline{g}_0(z) \\ &+ \underline{g}_0(z) \cdot \underline{\sigma}(z) \cdot \underline{g}_0(z) \cdot \underline{\sigma}(z) \cdot \underline{g}_0(z) + \dots \end{aligned} \quad (\text{B5})$$

Dyson's equation can be written as a functional-matrix equation:

$$\underline{F}(\underline{g}) \equiv \underline{F} = \underline{g}_0 + \underline{g}_0 \cdot \underline{\sigma} \cdot \underline{g} - \underline{g} = \underline{0} \quad (\text{B6})$$

and the problem is to obtain a solution of this equation for the unknown matrix $\underline{g}$. If we start with a trial solution $\underline{g}^t$ and expand $\underline{F}$ about this solution and retain only the first-order terms then we will obtain

$$\underline{F}(\underline{g}) = \underline{F}(\underline{g}^t) + \left[\frac{\delta \underline{F}(\underline{g})}{\delta \underline{g}} \right]_{\underline{g}^t} \cdot (\underline{g} - \underline{g}^t) = \underline{0}. \quad (\text{B7})$$

Equation (B7) can be immediately solved to give an approximate result $\underline{g}^a$,

$$\underline{g}^a = \underline{g}^t - \left[\frac{\delta \underline{F}}{\delta \underline{g}} \right]_{\underline{g}^t}^{-1} \cdot \underline{F}(\underline{g}^t). \quad (\text{B8})$$

The matrix inverse required in Eq. (B8) can be easily found:

$$(\delta \underline{F} | \delta \underline{g})^{-1} = -(\underline{1} + \underline{g} \cdot \underline{\sigma}). \quad (\text{B9})$$

Since Eq. (B9) contains the unknown $\underline{g}$ which must be replaced by $\underline{g}^t$ that appears in Eq. (B8), we replace it by another trial function $\underline{g}^r$ and then substitute the result in Eq. (B8), thereby producing another approximation which we call $\underline{g}^v$:

$$\underline{g}^v = \underline{g}_0 + \underline{g}_0 \cdot \underline{\sigma} \cdot \underline{g}^t + \underline{g}^r \cdot \underline{\sigma} \cdot \underline{g}_0 + \underline{g}^r \cdot \underline{R} \cdot \underline{g}^t, \quad (\text{B10})$$

where $\underline{R} \equiv \underline{\sigma} \cdot \underline{g}_0 \cdot \underline{\sigma} - \underline{\sigma}$.

The matrix $\underline{g}^v$ is stationary with respect to variations in both $\underline{g}^t$ and $\underline{g}^r$ about the exact solution $\underline{g}$. A very convenient set of trial functions for the problem may be generated by taking linear combinations of the terms arising in the iterative expansion of the Dyson equation, Eq. (B6):

$$\underline{g}^t = \sum_{L=1}^N c_L \underline{T}^{(L)}, \quad \underline{g}^r = \sum_{L=1}^N d_L \underline{T}^{(L)}, \quad (\text{B11})$$

where

$$\begin{aligned} \underline{T}^{(1)} &= \underline{g}_0, \\ \underline{T}^{(2)} &= \underline{g}_0 \cdot \underline{\sigma} \cdot \underline{g}_0, \\ \underline{T}^{(3)} &= \underline{g}_0 \cdot \underline{\sigma} \cdot \underline{g}_0 \cdot \underline{\sigma} \cdot \underline{g}_0, \\ &\dots \end{aligned} \quad (\text{B12})$$

As a first approximation we select one-parameter trial functions:

$$\underline{g}^t = c_1 \underline{g}_0, \quad \underline{g}^r = d_1 \underline{g}_0. \quad (\text{B13})$$

Inserting these trial functions into Eq. (B10), minimizing with respect to c_1 and d_1 , and then solving the two linear equations for c_1 and d_1 we obtain the "best" matrix $\underline{g}$. A typical element is

$$\underline{g}_{jlmn}^{(z)} = \frac{\delta_{j,m} \delta_{l,n}}{z - i\hbar^{-1}\omega_{jl} - \sigma_{jlmn}(z)}. \quad (\text{B14})$$

The function $\underline{\sigma}$ is defined in Eq. (38) and requires an explicit form for $(\underline{G} | (z - i\hat{L}_1)^{-1} | \underline{G})$ and $\underline{\alpha}$. These two are defined in Eqs. (43) and (41), respectively. Thus the explicit form of $\underline{\sigma}$ can be obtained and from it the matrix $\underline{g}$ via Eq. (B14).

APPENDIX C

In this appendix we present the necessary details to obtain the imaginary part of the Green's function $\underline{G}(z)$ for real frequencies. For simplicity Eq. (48) can be symbolically rewritten as

$$2\pi G_{jlmn}(i\omega \pm \eta) := \left[\omega \mp i\eta - A_1 + \frac{A_2}{\omega \mp i\eta + A_3} \right]^{-1}, \quad (C1)$$

where we have used $z = i\omega \pm \eta$. Since the quantity A_3 is real, we must use the standard equation to properly deal with singularities:

$$\frac{1}{x \pm i\eta} = P \left[\frac{1}{x} \right] \mp i\pi\delta(x). \quad (C2)$$

Then Eq. (C1) becomes

$$\begin{aligned} \tilde{G}_{jlmn}(\omega \mp i\eta) := & \frac{1}{2\pi} \left[\omega \mp i\eta - A_1 + A_2 P \left[\frac{1}{\omega + A_3} \right] \right. \\ & \left. \pm i\pi A_2 \delta(\omega + A_3) \right]^{-1}, \end{aligned} \quad (C3)$$

where we have used the definition $\tilde{G}(i\omega \pm \eta) = G(i(\omega \mp i\eta)) \equiv \tilde{G}(\omega \mp i\eta)$. Since the equation contains a finite imaginary part in the form of the term $i\pi A_2 \delta(\omega + A_3)$ we can now set $\eta = 0$ without impunity as there is no danger of encountering singularities for real ω . We now have

$$\tilde{G}_{jlmn}(\omega) := \frac{1}{2\pi} [\omega - A_1 + S(\omega) \pm i\Gamma(\omega)]^{-1}. \quad (C4)$$

Here, $S(\omega) = A_2 P[(\omega + A_3)^{-1}]$ becomes a frequency-dependent line shift and $\Gamma(\omega) = \pi A_2 \delta(\omega + A_3)$ the frequency-dependent line broadening. The imaginary part

$$\begin{aligned} \sigma_{jlmn}^{(1)}(\omega) &= \sum_{p,r} P \left[\frac{\rho_r(n_l - n_p) - \bar{n}_l n_p}{\omega + \hbar^{-1}\omega_{p,l} - \hbar^{-1}\Omega_r} \right] \pm i\pi \sum_{p,r} [\rho_r(n_l - n_p) - \bar{n}_l n_p] \delta(\omega + \hbar^{-1}\omega_{p,l} - \hbar^{-1}\Omega_r) \\ &\equiv S_{jlmn}^{(1)}(\omega) \pm i\pi \Gamma_{jlmn}(\omega). \end{aligned} \quad (D2)$$

Here, $S^{(1)}(\omega)$ is the contribution to the line shift and $\Gamma^{(1)}(\omega)$ is the contribution to the linewidth from $\sigma^{(1)}$ and consequently $S(\omega) = \sum_{Q=1}^8 S^{(Q)}(\omega)$ and $\Gamma(\omega) = \sum_{Q=1}^8 \Gamma^{(Q)}(\omega)$. The line shift will be ignored in the present calculations as the resonance frequencies are not known *a priori* but are equated to the experimental values.

To evaluate the linewidth we convert the summation over r into an integral since the heat bath is much larger than the cell itself:

$$\Gamma_{jlmn}^{(1)}(\omega) = \hbar \delta_{j,m} \delta_{l,n} \sum_p \int dr w(r) [\rho(\Omega_r)(n_l - n_p) - \bar{n}_l n_p] \delta(\hbar\omega + \omega_p - \omega_l - \Omega_r), \quad (D3)$$

where $w(r)$ is a density of phonon states for which we use the Debye model,

$$w(r) = \frac{V\hbar}{2\pi^2} r^2. \quad (D4)$$

The quantity Ω_r is the bath phonon frequency,

$$\Omega_r = \nu r, \quad (D5)$$

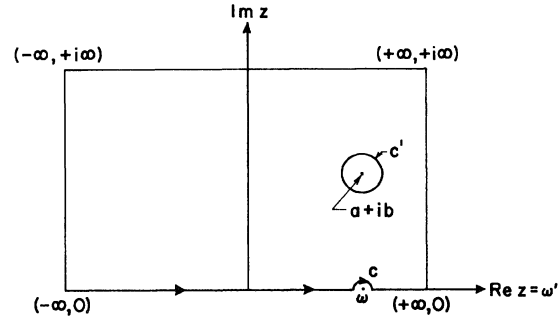


FIG. 1. Details of the contour used in Eq. (E6).

of Eq. (C4) is readily taken to give

$$\text{Im} \tilde{G}_{jlmn}(\omega) := \frac{1}{2\pi} \frac{\Gamma(\omega)}{[\omega - A_1 + S(\omega)]^2 + \Gamma^2(\omega)}. \quad (C5)$$

In all future development we will call $\tilde{G}(\omega)$ as simply $G(\omega)$.

APPENDIX D

The calculation of the linewidth function consists, essentially, of computing the summations contained in the eight terms in the bracket of Eq. (48) for real values of the frequency. Since the calculations with the eight different terms are repetitive, it is sufficient to consider just one term, e.g., $\sigma^{(1)}$, for illustrating the technique.

We begin with

$$\sigma_{jlmn}^{(1)}(z) = \sum_{p,r} \frac{\rho_r(n_l - n_p) - \bar{n}_l n_p}{z + i\hbar^{-1}\omega_{p,l} - i\hbar^{-1}\Omega_r}. \quad (D1)$$

Employing the discussion of Appendix C we obtain for real frequencies

where v is the speed of sound in the bath. Inserting Eqs. (D4) and (D5) into Eq. (D3) gives

$$\Gamma_{jlmn}^{(1)}(\omega) = \frac{V\hbar}{2\pi^2 v^3} \sum_p (\hbar\omega + \omega_p - \omega_l)^2 [\rho(\hbar\omega + \omega_p - \omega_l)(n_l - n_p) - \bar{n}_l n_p] \delta_{j,m} \delta_{l,n}. \quad (D6)$$

From the discussion given in Sec. V we can see that since $\sigma^{(1)}$ is a term of category (b), the final summation in Eq. (D6) is approximated by a single term $p = k$ and

$$\Gamma_{jlmn}^{(1)}(\omega) = \frac{V\hbar}{2\pi^2 v^3} (\hbar\omega + \omega_k - \omega_l)^2 [\rho(\hbar\omega + \omega_k - \omega_l)(n_l - n_k) - \bar{n}_l n_k] \delta_{j,m} \delta_{l,n}. \quad (D7)$$

APPENDIX E

In this appendix we present some of the mathematical details required for the calculation of the principal-value integration appearing in Eq. (14). In the high-temperature limit ($\beta \rightarrow 0$) the ratio of the two exponential factors in Eq. (53) simplifies to

$$\epsilon''(\omega) = \frac{2\pi^2 \omega}{VkT} \sum_{j,l} \frac{|\langle j | \hat{\mu} | l \rangle|^2 (|\chi|^2 / \pi^2 \hbar^2) \Gamma_{jl}(\hbar^{-1} \omega_{j,l})}{(\omega - \hbar^{-1} \omega_{j,l})^2 + [|\chi|^4 / \pi^2 \hbar^4 (n_l - n_j)^2] \Gamma_{jl}^2(\hbar^{-1} \omega_{j,l})}. \quad (E1)$$

Substituting Eq. (E1) into Eq. (14) we obtain

$$\epsilon'(\omega) = 1 + \frac{2|\chi|^2}{VkT\hbar^2} \sum_{j,l} |\langle j | \hat{\mu} | l \rangle|^2 \Gamma_{jl}(\hbar^{-1} \omega_{j,l}) P \int_{-\infty}^{\infty} dv \frac{v}{\{(\omega - \hbar^{-1} \omega_{j,l})^2 + [|\chi|^4 / \pi^2 \hbar^4 (n_l - n_j)^2] \Gamma_{jl}^2(\hbar^{-1} \omega_{j,l})\} (v - \omega)}. \quad (E2)$$

We define

$$a \equiv \hbar^{-1} \omega_{j,l}, \quad (E3)$$

$$b^2 \equiv \frac{|\chi|^4}{\pi^2 \hbar^4 (n_l - n_j)^2} \Gamma_{jl}^2(\hbar^{-1} \omega_{j,l}). \quad (E4)$$

The principal-value integral can be given in a shorthand form by

$$I \equiv P \int_{-\infty}^{\infty} dv \frac{v}{[(v-a)^2 + b^2](v-\omega)}. \quad (E5)$$

Instead of I as defined by Eq. (E5) we consider another integral J ,

$$J \equiv \int dz \frac{z}{(z-a+ib)(z-a-ib)(z-\omega)}. \quad (E6)$$

Here, the contour of integration over the variable z is selected according to Fig. 1. It is easy to show that the integrand vanishes along the contour except along the lines $-\infty < \chi \leq \omega - i\eta$ and $\omega + i\eta \leq \chi \leq +\infty$ and the infinitesimally small semicircular contour C . Since the integrand is analytic over the entire upper half of the com-

plex plane, with the exception of the point $z = a + ib$, the integral can be evaluated by shrinking the large, almost rectangular contour to an infinitesimally small circular contour around the simple pole at $(a + ib)$. Thus

$$\begin{aligned} P \int_{-\infty}^{\infty} dx \frac{x}{(x-a+ib)(x-a-ib)(x-\omega)} \\ + \int_C dz \frac{z}{(z-a+ib)(z-a-ib)(z-\omega)} \\ = \oint_{C'} dz \frac{z}{(z-a+ib)(z-a-ib)(z-\omega)}. \end{aligned} \quad (E7)$$

The right-hand side is evaluated by applying Cauchy's theorem as $\pi(a+ib)/b(a+ib-\omega)$. The second term in the left-hand side is calculated by changing to an angular variable θ , $z = \omega + \eta e^{i\theta}$, where θ extends from π to 0 and η is the radius of the semicircle C . Upon integration over θ we get for this term $-\pi\omega i / [(\omega-a+ib)(\omega-a-ib)]$. Collecting all the results yields

$$I = \frac{\pi(a^2 + b^2 - a\omega)}{b[(\omega-a)^2 + b^2]}. \quad (E8)$$

¹H. Fröhlich, Phys. Lett. **26A**, 402 (1968).

²H. Fröhlich, Int. J. Quantum Chem. **2**, 641 (1968).

³H. Fröhlich, Nature (London) **228**, 1093 (1970).

⁴H. Fröhlich, Phys. Lett. **39A**, 113 (1972).

⁵H. Fröhlich, Phys. Lett. **51A**, 21 (1975).

⁶H. Fröhlich, Nuovo Cimento **7**, 416 (1977).

⁷H. Fröhlich, Riv. Nuovo Cimento **7**, 399 (1977).

⁸T. M. Wu and S. Austin, Phys. Lett. **64A**, 151 (1977).

⁹T. M. Wu and S. Austin, Phys. Lett. **65A**, 74 (1978).

¹⁰T. M. Wu and S. Austin, Phys. Lett. **73A**, 166 (1979).

¹¹T. M. Wu and S. Austin, J. Theor. Biol. **71**, 209 (1978).

¹²R. K. Mishra, K. Bhaumik, A. Srivastava, and S. S. Chaudhary, Int. J. Quantum Chem. **20**, 377 (1981).

¹³D. A. McQuarrie, *Statistical Mechanics* (Harper and Row, New York, 1976).

¹⁴R. London, *The Quantum Theory of Light* (Clarendon, Oxford, 1973).

¹⁵R. Paul, *Field Theoretical Methods in Chemical Physics* (Elsevier Science, Amsterdam, 1982).

¹⁶J. B. Hasted, *Aqueous Dielectrics* (Chapman and Hall, London, 1973).